Responsibly Deploy AI in Quantitative Social Research

In-class demo for NMC5344 Coding for Communicators

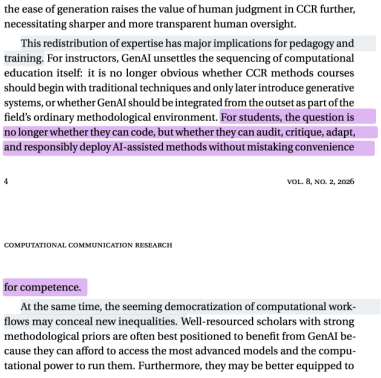
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Email: rongxin@u.nus.edu / Website: https://rongxin.me
Using causal and computational methods to evaluate
societal and political impacts of digital information technologies

November 7, 2026; NUS

The goal of this session

... lower barrier (of computational skills)
... redistribution of expertise ...
"For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience..." (Zhou et al., 2026)



the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

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COMPUTATIONAL COMMUNICATION RESEARCH

for competence.

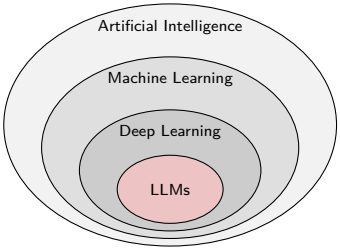
At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

Source: Zhou et al. (2026)

Introduction

1.1 What is artificial intelligence?

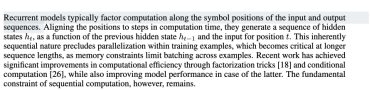
- Computational agents that act intelligently (Poole & Mackworth, 2023);
- But what is intelligence? See the figure
- More importantly, please be aware of the differences ...



1.2 A simplified history of LLMs

- * Artificial intelligence is an old field developed decades ago (see Poole & Mackworth, 2023);
- * Pre-LLMs: early tools mainly classified, clustered, or predicted from human-produced traces (Zhou et al., 2026). [a]
- * Birth of LLMs: the Transformer (Vaswani et al., 2017), a model architecture that parallelizes training, enabling large datasets and model sizes
- Post-LLMs: we focus on the LLM era after 2020 (marked by GPT-3; Xu et al., 2026) and its implications for computational social science research

[a] Likewise, Percy Liang uses 2016, 2018, and today to divide three time periods; see https://cs336.stanford.edu/lectures/?trace=lecture_01



Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

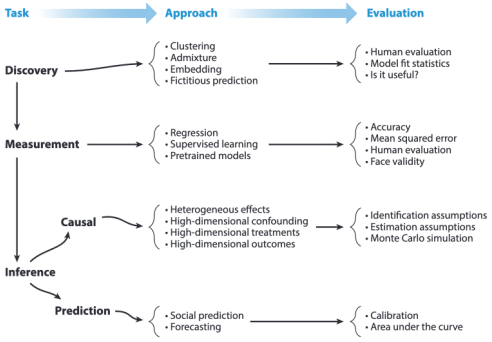
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

Excerpt from Vaswani et al. (2017)

Pre-LLMs AI

2.1 The use of AI in social science before LLMs

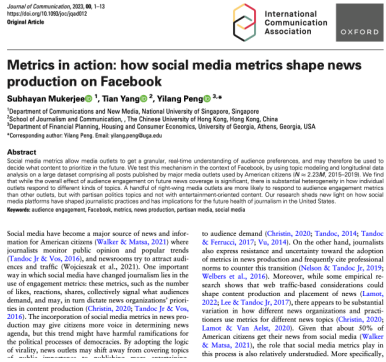
- Social science research often leverages machine learning for various tasks, according to the typology in Grimmer et al. (2021):
  - Discovery; e.g., text classification (Tutorial I)
  - Measurement; e.g., regression
  - Causal Inference; e.g., heterogeneous treatment effects
  - Prediction; e.g., forecasting social trends



Source: Grimmer et al. (2021) (p.23)

2.2 Example: Metrics in action

- Mukerjee et al. (2023) ask how audience engagement metrics shape news organizations' content production on Facebook
- Topic modeling (discovery) on all Facebook posts by major U.S. media outlets, combined with longitudinal analysis



Journal of Communication, 2023, 00, 1–13
https://doi.org/10.1093/joc/jqad012
Original Article

International Communication Association

OXFORD

Metrics in action: how social media metrics shape news production on Facebook

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Abstract

Social media metrics allow media outlets to get a granular, real-time understanding of audience preferences, and may therefore be used to decide what content to prioritize in the future. We test this mechanism in the context of Facebook, by using topic modeling and longitudinal data analysis on a large dataset comprising all posts published by major media outlets used by American citizens ($N = 2.22$M, 2015–2018). We find that while the overall effect of audience engagement on future news coverage is significant, there is substantial heterogeneity in how individual outlets respond to different kinds of topics. A handful of right-wing media outlets are more likely to respond to audience engagement metrics than other outlets, but with partisan politics topics and not with entertainment-oriented content. Our research sheds new light on how social media platforms have shaped journalistic practices and has implications for the future health of journalism in the United States.

Keywords: audience engagement, Facebook, metrics, news production, partisan media, social media

Social media have become a major source of news and information for American citizens (Walker & Matsa, 2021) where journalists monitor public opinion and popular trends (Tandoc Jr & Vos, 2016), and newsrooms try to attract audiences and traffic (Wojcieszak et al., 2021). One important way in which social media have changed journalism lies in the use of engagement metrics: these metrics, such as the number of likes, reactions, shares, collectively signal what audiences demand, and may, in turn dictate news organizations' priorities in content production (Christin, 2020; Tandoc Jr & Vos, 2016). The incorporation of social media metrics in news production may give citizens more voice in determining news agenda, but this trend might have harmful ramifications for the political processes of democracies. By adopting the logic of virality, news outlets may shift away from covering topics to audience demand (Christin, 2020; Tandoc, 2014; Tandoc & Ferrucci, 2017; Vu, 2014). On the other hand, journalists also express resistance and uncertainty toward the adoption of metrics in news production and frequently cite professional norms to counter this transition (Nelson & Tandoc Jr, 2019; Welbers et al., 2016). Moreover, while some empirical research shows that web traffic-based considerations could shape content production and placement of news (Lamot, 2022; Lee & Tandoc Jr, 2017), there appears to be substantial variation in how different news organizations and practitioners use metrics for different news topics (Christin, 2020; Lamot & Van Aelst, 2020). Given that about 50% of American citizens get their news from social media (Walker & Matsa, 2021), the role that social media metrics play in this process is also relatively understudied. More specifically,

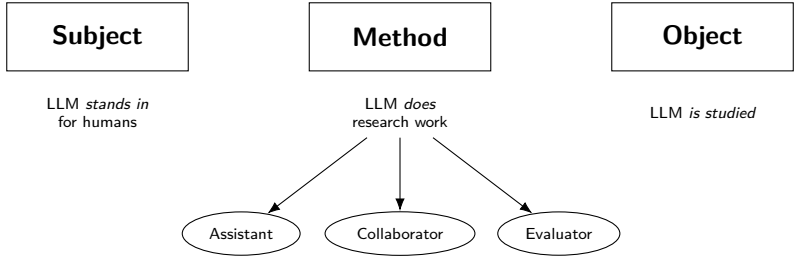
Downloaded from https://academic.oup.com/joc/advance-article/doi/10.1093/joc/jqad012/7138729 by [Chine

Source: Mukerjee et al. (2023)

LLMs

3.1 Roles of LLMs in Computational Social Science

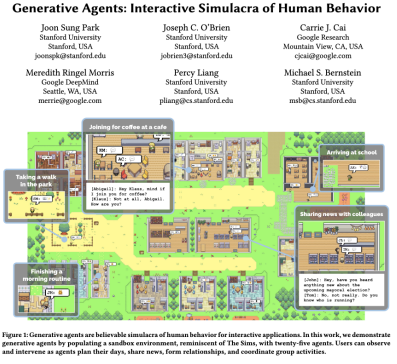
- Drawing on more perspectives, we propose a comprehensive categorization of LLM roles in CSS, elaborated in the following slides



Source: Four roles (subjects, assistants, collaborators, evaluators) categorized in Xu et al. (2026); four categories in H. Liang (2026); and two roles (a research tool, an object of inquiry) adapted from Zhou et al. (2026)

3.2 LLMs as Subjects: Categories

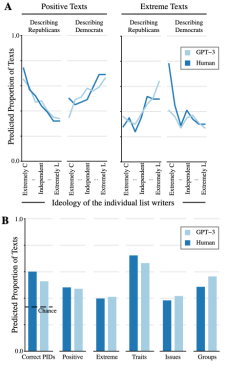
- LLMs serve as research subjects, according to Xu et al. (2026):
  - Generating survey responses (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - Simulating behavior sequences (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - Simulating social interactions (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



Generative agents in a simulated town; Park, O'Brien, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

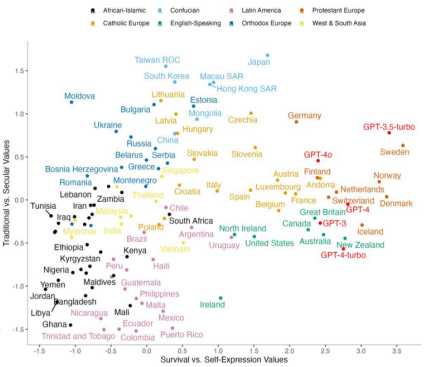
- **Pros:**
  - Successfully replicate human responses in surveys (Argyle, Busby, et al., 2023)
  - Produce believable individual and emergent social behaviors (Park, O'Brien, et al., 2023)
  - Replicate emergence of social conventions (Ashery et al., 2025)
  - etc.



Analysis of GPT-3 and human responses from the Lucid survey; Fig 3 in Argyle, Busby, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

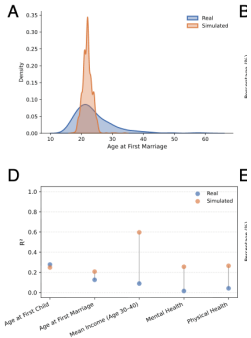
- **Cons:**
  - Poorly reflect "hard-to-reach populations" (Bollen et al., 2026); fail to capture experiential and cultural depth (Pataranutaporn et al., 2025)
  - Reflect certain partisan (Santurkar et al., 2023) or cultural biases (Tao et al., 2024)
  - Close to averages but with less variation (Bisbee et al., 2024); have strong, varying biases and have low variance in opinion or attitudinal research (Boelaert et al., 2025); likely to both misportray and flatten demographic representations (A. Wang et al., 2025); and tend to compress heterogeneity into simplified typological structures (Y. Xie & Xie, 2025)



LLM-generated answers on values deviate from those of different countries; Fig 1 in Tao et al. (2024)

3.3 LLMs as Subjects: Pros and Cons

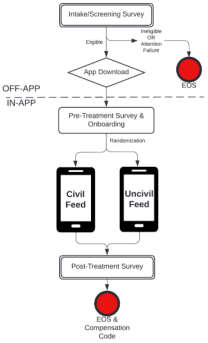
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Failing to capture the dispersion of real-world distributions; Fig 3 in Y. Xie & Xie (2025)

3.4 LLMs as Methods: Assistant

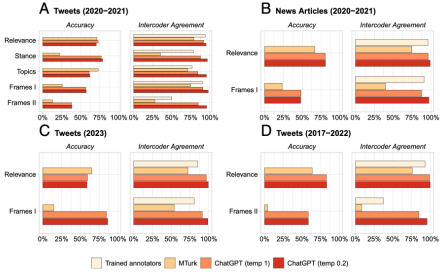
- Interviewing human participants (qualitative) (Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **Supporting experiments** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- Analysing textual content (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Synthetic communicative environment in a randomized experiment; Allamong et al. (2026)

3.4 LLMs as Methods: Assistant

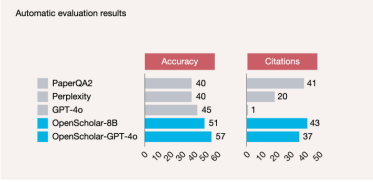
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- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Better annotation than crowd-workers; Gilardi et al. (2023)

3.5 LLMs as Methods: Collaborator

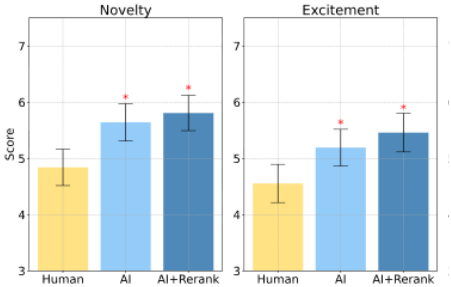
- **Reviewing literature** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- Proposing hypotheses (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)



PhD-level literature review system powered by LLMs; see Asai et al. (2026)

3.5 LLMs as Methods: Collaborator

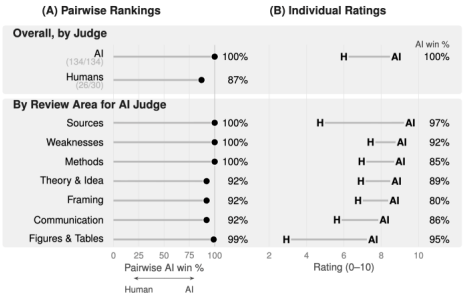
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- **Proposing hypotheses** (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)



Greater novelty and excitement in LLM-generated hypotheses; see Si et al. (2024)

3.6 LLMs as Methods: Evaluator

- Evaluating research protocols (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **Critiquing research outcomes** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; W. Liang et al., 2024)



AI reviews dominate human reviews; source: (King & Worthington, 2026); not peer-reviewed but well designed

3.7 LLMs as Objects

- "A communication technology that reconfigures production, circulation, interpretation, and power across media systems, raising critical questions about authenticity, access, and the reproduction of social inequalities." (Zhou et al., 2026)
- For example, Liu and Zhang (2026) evaluates how LLM-generated summaries shape public sentiment, where the treatment is the model itself.

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

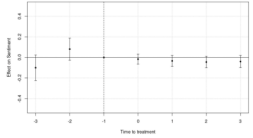


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

AIGS as objects in Liu & Zhang (2026)

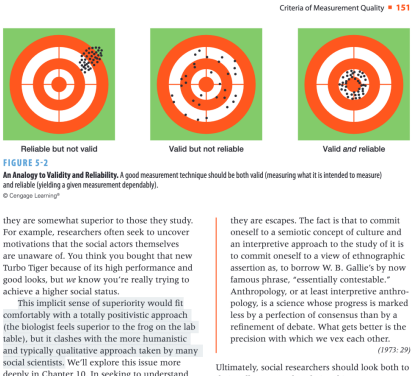
TUT

Agentic Modelling in Data Analysis
Online material: GitHub

Deploy Responsively

4.1 LLMs deployed responsibly: Validity and Reliability

- Again, always verify the validity and reliability of the model outputs before using them in your research



Criteria of Measurement Quality ■ 151

Reliable but not valid | Valid but not reliable | Valid *and* reliable

FIGURE 5-2

An Analogy to Validity and Reliability. A good measurement technique should be both valid (measuring what it is intended to measure) and reliable (yielding a given measurement dependably).

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they are somewhat superior to those they study. For example, researchers often seek to uncover motivations that the social actors themselves are unaware of. You think you bought that new Turbo Tiger because of its high performance and good looks, but *we* know you're really trying to achieve a higher social status.

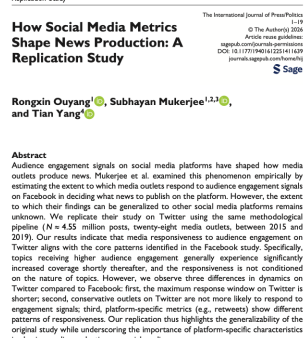
This implicit sense of superiority would fit comfortably with a totally positivistic approach (the biologist feels superior to the frog on the lab table), but it clashes with the more humanistic and typically qualitative approach taken by many social scientists. We'll explore this issue more deeply in Chapter 10. In seeking to understand they are escapes. The fact is that to commit oneself to a semiotic concept of culture and an interpretive approach to the study of it is to commit oneself to a view of ethnographic assertion as, to borrow W. B. Gallie's by now famous phrase, "essentially contestable." Anthropology, or at least interpretive anthropology, is a science whose progress is marked less by a perfection of consensus than by a refinement of debate. What gets better is the precision with which we vex each other. *(1973: 29)*

Ultimately, social researchers should look both to

Validity and reliability; from Babbie (2021)

4.2 LLMs deployed responsibly: Reproducibility

- Reproducibility is important to science
- Ensure that the AI part of your research is reproducible by:
  - Preferring API-based access over chat-based access
  - Documenting model versions, prompts, thinking level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the random seed and temperature (search for details)



Replication Study

The International Journal of Press/Politics
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How Social Media Metrics Shape News Production: A Replication Study

Rongxin Ouyang[1], Subhayan Mukerjee[1,2,3], and Tian Yang[4]

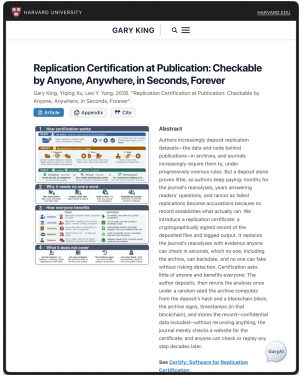
Abstract

Audience engagement signals on social media platforms have shaped how media outlets produce news. Mukerjee et al. examined this phenomenon empirically by estimating the extent to which media outlets respond to audience engagement signals on Facebook in deciding what news to publish on the platform. However, the extent to which their findings can be generalized to other social media platforms remains unknown. We replicate their study on Twitter using the same methodological pipeline ($N \approx 4.55$ million posts, twenty-eight media outlets, between 2015 and 2019). Our results indicate that media responsiveness to audience engagement on Twitter aligns with the core patterns identified in the Facebook study. Specifically, topics receiving higher audience engagement generally experience significantly increased coverage shortly thereafter, and the responsiveness is not conditioned on the nature of topics. However, we observe three differences in dynamics on Twitter compared to Facebook: first, the maximum response window on Twitter is shorter; second, conservative outlets on Twitter are not more likely to respond to engagement signals; third, platform-specific metrics (e.g., retweets) show different patterns of responsiveness. Our replication thus highlights the generalizability of the original study while underscoring the importance of platform-specific characteristics in shaping media production on social media.

Replicating Mukerjee et al. (2023) on Twitter; Source: Ouyang et al. (2026)

4.2 LLMs deployed responsibly: Reproducibility

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  - Documenting model versions, prompts, thinking level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the random seed and temperature (search for details)



Replication certification at publication; Source: King et al. (2026)

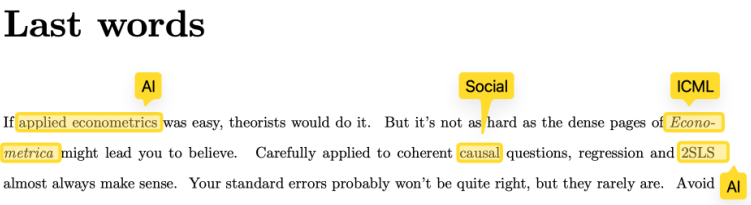
Takeaways

5.1 Takeaways

- AI models (including LLMs) are just tools for high-dimensional data analysis; they can be subjects, objects, and methods in quantitative social research
- As methods, they are expected to be used in a responsible way in which validity, reliability, and reproducibility are carefully considered
- Like any other methods, although they evolve rapidly, their use should rest on our taste for good questions in research, industry, and life, which remains central to higher education.

5.1 Takeaways

- "... Be skeptic... don't panic!" (Angrist & Pischke, 2009)



Last words

AI Social ICML

If applied econometrics was easy, theorists would do it. But it's not as hard as the dense pages of *Econometrica* might lead you to believe. Carefully applied to coherent causal questions, regression and 2SLS almost always make sense. Your standard errors probably won't be quite right, but they rarely are. Avoid embarrassment by being your own best skeptic - and, especially, Don't Panic!

AI

Source: Angrist & Pischke (2009)

Resources

6.1 Resources

- **Locate the most appropriate AI model for your task**
  - Artificial Analysis https://artificialanalysis.ai/
  - Arena.ai, blind benchmarking for AI models https://arena.ai/leaderboard/
- **Find problems in your manuscript**
  - Refine, publication-level technical review; adopted by AEA https://www.refine.ink/
  - OpenAIReview, open-source multi-agent manuscript review solution by UChicago HAI https://github.com/ChicagoHAI/OpenAIReview
- **Double-check references and citations**
  - CiteVerify, "a local tool for verifying references in research papers", https://github.com/cuihuash/CiteVerify
  - Chatting with frontier models also helps

6.1 Resources

- **Organize multiple AI models and services**
  - For macOS, magpie, a local AI router manager for Mac https://github.com/yetone/magpie
  - For other platforms, cc-switch, https://github.com/farion1231/cc-switch
- **Skills**
  - AER-skills, https://github.com/brycewang-stanford/AER-skills
  - Claude Code setup by an Econ prof, https://psantanna.com/claude-code-my-workflow/
- **Read scientific literature in your own language**
  - PDFMathTranslate, read scientific PDFs in your language https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)
- **More examples if you find them helpful...**
  - 打造 AI 科学家: DeepSeek 与 ChatGPT 应用实战手册 (机械工业出版社, 2026)

Acknowledgments

Thanks to Hongzhan Lin, Yanlin Li, Hao Li, Ziqi Lu, Long Wang, Bei Zhan, Jianjun Yu, and Qinghua Qin for exchanging early ideas on this topic with me. All errors are my own.

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